

Yingjing Huang

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Research Interest

GIScience, GeoAI, Deep Learning, Computer Vision, Urban Visual Intelligence, Social Sensing

Education

Peking University

Ph.D. in Geographic Information Science

• Advisors: Yu Liu, Fan Zhang

Beijing, China

Sep. 2021 - Jul. 2025)

Wuhan University

MS. in Geographic Information Science

• Advisor: Teng Fei

Wuhan, China

Sep. 2018 - Jun. 2021

Wuhan University

BSc. in Geographic Science

• Advisors: Yiyun Chen, Teng Fei

Wuhan, China

Sep. 2013 - Jun. 2017

Experience

MIT Senseable City Lab

Visiting Ph.D. Student

• Advisors: Carlo Ratti, Fábio Duarte, Simone Mora

Cambridge, US

Jan. 2023 - Jan. 2024

Publications

Publications reported in reverse chronological order

(# co-first author, Google citations = 180)

Ongoing and Submitted Manuscripts

- [1] **Yingjing Huang**[#], Yong Li, Fan Zhang, & Yu Liu. 2025. Measuring sustainable development with urban imagery. (In preparing)
- [2] **Yingjing Huang**, Fan Zhang, & Yanhua Yao. 2025. Quantifying inequality and imbalance of urban indoor and outdoor space quality. (In preparing)
- [3] Yong Li, **Yingjing Huang**[#], Gengchen Mai, Fan Zhang. 2025. Learning street view representations with spatiotemporal contrast. *Computers, Environment and Urban Systems*. (Under Review, arxiv:2502.04638)

Peer-reviewed Journal Articles

- [1] **Yingjing Huang**, Fan Zhang, Lun Wu, and Yu Liu. 2025. Measuring urban physical environments using image deep features. *Cities*, 166, 106196. (IF = 6.6, Rank in Urban Studies = 7/288, Q1)
- [2] **Yingjing Huang**, Fan Zhang, Yong Li, Lun Wu, & Yu Liu. 2025. Intelligent computational representation of urban imagery. *Geomatics and Information Science of Wuhan University*. (In Chinese a leading journal in Geomatics in China, indexed by EI Compendex)
- [3] **Yingjing Huang**, Rohit Sanatani, Chang Liu, Yuhao Kang, Fan Zhang, Yu Liu, Fabio Duarte, and Carlo Ratti. (2025). No “true” greenery: Deciphering the bias of satellite and street view imagery in urban greenery measurement. *Building and Environment*, 269, 112395. (IF = 7.6, Rank in Geography, Planning and Development = 18/841, Q1)
- [4] **Yingjing Huang**, Fan Zhang, Yong Gao, Wei Tu, Fabio Duarte, Carlo Ratti, Diansheng Guo, and Yu Liu. 2023. Comprehensive urban space representation with varying numbers of street-level images.

Computers, Environment and Urban Systems, 106, 102043. (IF = 8.3, Rank in Urban Studies = 3/288, Q1)

- [5] Meng Bian, Shuyi Guo, Wei Wang, Yuhui Ouyang, **Yingjing Huang**, and Teng Fei. 2021. Nextday forecast of Beijing pollen concentration fused with vegetation remote sensing data—Implementing a nonlinear autoregressive neural network model with external input. *Journal of Geo-Information Science*, 2021,23(09):1705-1713. (in Chinese, a key academic journal in GIS, indexed in the CSCD Core database)
- [6] **Yingjing Huang**, Teng Fei, Mei-Po Kwan, Yuhao Kang, Jun Li, Yizhuo Li, Xiang Li, and Meng Bian. 2020. GIS-Based Emotional Computing: A Review of Quantitative Approaches to Measure the Emotion Layer of Human–Environment Relationships. *ISPRS International Journal of Geo Information*, 9, 551. (IF = 2.8, Rank in Geography, Planning and Development = 95/841, Q2)
- [7] **Yingjing Huang**, Jun Li, Guofeng Wu and Teng Fei. 2020. Quantifying the bias in place emotion extracted from photos on social networking sites: A case study on a university campus. *Cities*, 102, 102719. (IF = 6.6, Rank in Urban Studies = 7/288, Q1)
- [8] Yizhuo Li, Teng Fei, **Yingjing Huang**, Jun Li, Xiang Li, Fan Zhang, Yuhao Kang and Guofeng Wu. 2020. Emotional habitat: mapping the global geographic distribution of human emotion with physical environmental factors using a species distribution model. *International Journal of Geographical Information Science*, 1-23. (IF = 5.1, Rank in Geography, Planning and Development = 31/841, Q1)
- [9] Shuangyin Zhang, Jun Li, Siying Wang, **Yingjing Huang**, Yizhuo Li, Yiyun Chen and Teng Fei. 2020. Repaid Identification and Prediction of Cadmium–Lead Cross-Stress of Different Stress Levels in Rice Canopy Based on Visible and Near-Infrared Spectroscopy. *Remote Sensing*, 12, 469. (IF = 4.1, Rank in General Earth and Planetary Sciences = 13/198, Q1)
- [10] Yiyun Chen, Tianci Qi, **Yingjing Huang**, Yuan Wan, Ruiying Zhao, Lin Qi, Chao Zhang, and Teng Fei. 2017. Optimization method of calibration dataset for VIS-NIR spectral inversion model of soil organic matter content. *Transactions of the Chinese Society of Agricultural Engineering*, 06, 107-114. (in Chinese, a top-tier journal in Agricultural Engineering in China, indexed by EI Compindex)

Patents

- [1] Fan Zhang, **Yingjing Huang**, Yu Liu. Method, Device, and Storage Medium for Comprehensive Landscape Representation Based on Street Imagery (in Chinese, National Invention Patent, China, Received, Not Yet Published)
- [2] Fan Zhang, **Yingjing Huang**, Diansheng Guo. Image Processing Method, Device, Product, Equipment, and Medium (in Chinese, National Invention Patent, China, Published, Not Yet Granted)
- [3] Fan Zhang, **Yingjing Huang**, Diansheng Guo. Method and Related Device for Identifying Regions in Space (in Chinese, National Invention Patent, China, Published, Not Yet Granted)

Honors & Awards

2020	“Excellent graduate student” National award for the excellent graduate student in Wuhan University	Wuhan, China
2020	Kwang-Hua Scholarship Scholarship to the excellent student	Wuhan, China
2020	Second-Class Scholarship Awards to outstanding student in academic in Wuhan University	Wuhan, China
2015	Third-Class Scholarship Awards to outstanding student in academic in Wuhan University	Wuhan, China

Talks

Comprehensive Urban Space Representation with Varying Numbers of Street-Level Images

Honolulu, US

American Association of Geographers 2024 Annual Meeting

Apr. 2024

- Oral Presentation

Quantifying the bias in place emotion extracted from photos on social networking sites: A case study on a university campus

Beijing, China

2021 International Graduate Workshop on GeoInformatics

Dec. 2021

- Oral Presentation (online)

Community Services

Journal Reviewer

- Cities
- Computers, Environment and Urban Systems
- Landscape and Urban Planning
- Urban Forestry & Urban Greening
- Travel Behaviour and Society
- Applied Geography
- Scientific Reports
- IEEE Transactions on Human-Machine Systems
- Journal of Urban Technology
- Computational Urban Science
- Journal of Transportation Engineering and Information (in Chinese)

Grants

2025-2029	Key Program of the National Natural Science Foundation of China (No. 42430106) Spatio-temporal optimization with geographical artificial intelligence	Contributor
2024-2026	National Natural Science Foundation of China (No. 42371468) Representing Urban Physical Environment using Street View Imagery	Contributor
2022-2025	National Natural Science Foundation of China-Excellent Young Scientists Fund Program (Overseas) (No. 41901321) Geospatial Data Mining & Visual Intelligence	Contributor
2022-2023	International Research Center of Big Data for Sustainable Development Goals (CBAS) Key Technologies and Applications of Big Earth Data for Sustainable Development Goals	Contributor
2021-2022	Tencent Academic Research Fund (No. RAGR20210101) Identifying urban informal settlement using multi-source big geo-data	Key Contributor
2020-2022	National Natural Science Foundation of China (No. 41901321) Understanding the relationship between urban physical environment and urban mobility patterns quantitatively using street view imagery	Contributor
2017-2021	National Basic Research Program of China (No. 2017YFB0503602) Geospatial Big Data Mining and Spatiotemporal Pattern Discovery	Contributor

Skills

Programming	Python, Matlab, R, Javascript, LaTeX
Packages	Sklearn, Pytorch
Databases	MySQL, SQLite, MongoDB
GIS skills	ArcGIS, QGIS, Geopandas, Arcpy
OS	Windows, Linux, Mac OS
Languages	Mandarin Chinese (<i>Native</i>), English (<i>Advanced</i>)